Laboratory Exercise 01

Topics

1. Confidence with the IDE Visual Studio Code
2. Print function
3. Basic arithmetic operations

Exercises

Part 1 – Basic operations to start with Python

***Delivery**: For each of the following exercises, write a program in Python that responds to the requests indicated. Complete at least three exercises during the exercise, and the rest at home.*

01.1.1 Two numbers. Write a program that stores two integers in two constants defined in code, and then displays:

A. The sum;
B. The difference;
C. The product;
D. The average value;
E. The distance (i.e. the absolute value of the difference);
F. The maximum value (i.e. the greater of the two);
G. The minimum value (i.e. the lesser of the two).

```
##
# Save two integers in two constants, and display the result of several
calculations involving them.
#
NUM_1 = 20
NUM_2 = 40
# Alternatively, read the integers from the user.
# NUM_1 = int(input("Enter an integer: "))
# NUM_2 = int(input("Enter another integer: "))
# Compute and display the sum, difference, product, average, distance,
# minimum and maximum.

Sum = NUM_1 + NUM_2
Difference = NUM_1 - NUM_2
Product = NUM_1 * NUM_2
Average = (NUM_1 + NUM_2) / 2
Distance = abs(NUM_1 - NUM_2)
Min = min(NUM_1, NUM_2)
Max = max(NUM_1, NUM_2)

print("Sum:", Sum)
print("Difference:", Difference)
print("Product:", Product)
print("Average:", Average)
print("Distance:", Distance)
print("Minimum:", Min)
print("Maximum:", Max)
```

Tip: Use the `max()` and `min()` functions defined in Python. They accept a comma-separated sequence of input values and return the maximum and minimum values of the sequence, respectively (for example, `max(10, 5)` returns `10`). [P2.4]

01.1.2 Car's use. You want to find out which fraction of your car's use is for going to work, and which is for personal use. You know the one-way distance from your home to work. For a particular period, you recorded the beginning and ending kilometers on the odometer and the number of working days. Try to execute the follow code able of computing the fraction of car's use for work and personal. Use the debug during the analysis.

```
km_start = int(input("insert beginning kilometers: "))
km_stop = int(input("insert ending kilometers: "))
working_days = int(input("insert the number of working days: "))
km_home_work = int(input("insert the home-work discance: "))

km_total = km_stop - km_start
km_work = working_days * (km_home_work *2)
km_not_work = km_total - km_work

km_work = 100 * (km_work/km_total)
km_not_work = 100 * (km_not_work/km_total)

print(10*"-")
print("car used for work activities: ", km_work)
print("car used for not work activities: ", km_not_work)
```

01.1.3 Restaurant. Imagine that you and a number of friends go to a luxury restaurant, and when you ask for the bill, you want to split the total amount and the tip (15 percent) between all of you. Your program should print the total amount of the bill, the tip amount, the total cost (so the bill and the tip), and the amount each person has to pay. It should also print how much of what each person pays is for the bill and how much for the tip.

01.1.4 Figures. Write a program that prints the sum of the first ten positive integers, `1 + 2 + … + 10`. [P1.2]

01.1.5 Interests. Write a program that prints the balance of an account after the first, second, and third year. The account has an initial balance of $1,000 and earns 5 percent interest per year. [P1.4]

01.1.6 The "Game of Codes" Throne Contest. Arya and Jon are competing for the Iron Throne by collecting points. Store Arya's and Jon's points as constants (e.g., `ARYA_PT = 87`, `JON_PT = 92`).

- Print the maximum points.
- Print the absolute difference in their scores.
- Print the average score.
- (optional) Now try to rewrite the program asking for the points as inputs from the user. *Tip*: use the `input()` function with `int()` conversion.

01.1.7 The "Stranger Things" Starcourt Mall Snack Split. Eleven, Mike, and Dustin went to the Starcourt Mall. Eleven bought 3 ice creams, Mike bought 2, and Dustin bought 4. Each ice cream costs $5.

- Store the number of ice creams each person bought as constants.
- Calculate and print the total bill.
- Print how much each person should pay if they split the bill equally.